QUANTUM CRYPTOGRAPHY (BB84 PROTOCOL) — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on Quantum Cryptography (BB84 Protocol)

Question 1 10.0 points

Analyze the diagram below of the four BB84 quantum states on the Bloch sphere. Describe the positions of the rectilinear and diagonal bases and explain their significance in the BB84 protocol.

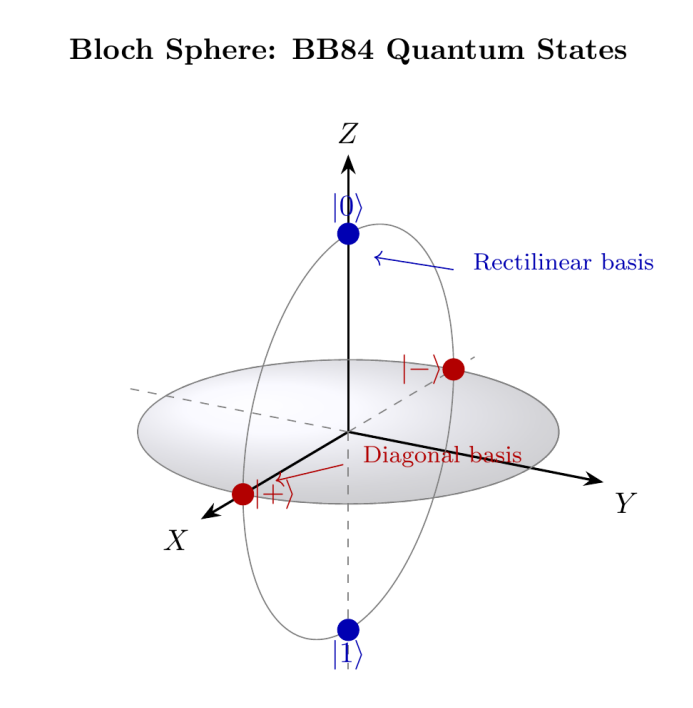

Fig. 1. Bloch sphere representation of BB84 quantum states. Show the Bloch sphere with axes labeled X, Y, and Z. Indicate the

Question 2

10.0 points

Evaluate the BB84 key exchange sequence in the diagram below, which shows an intercept-resend attack by Eve. Explain how the attack affects the key distribution between Alice and Bob.

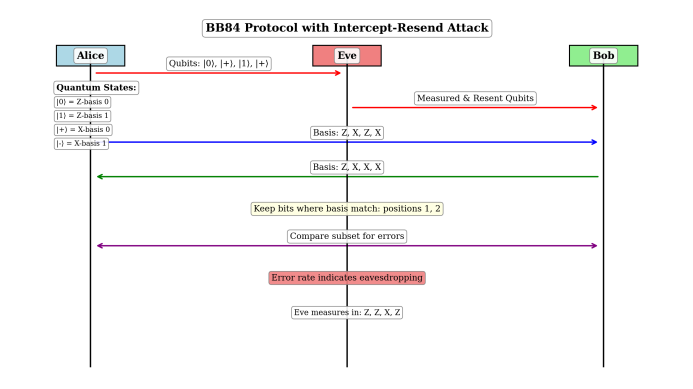

Fig. 2. BB84 key exchange sequence diagram showing an intercept-resend attack by an eavesdropper, Eve. The diagram should include the following components: Alice, Bob, and Eve as entities. Arrows representing qubit transmission from Alice to Bob, with Eve intercepting and resending qubits. Indicate the basis choices (rectilinear or diagonal) and measurement outcomes. Highlight the points where Eve's interference affects the key distribution process. Label all entities and key steps clearly to illustrate the sequence of the BB84 protocol and the impact of the attack. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Incorrect • eavesdropping The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Examine the quantum key distribution channel diagram with the eavesdropping detection probability graph shown below. Discuss how the graph informs Alice and Bob about potential eavesdropping activities.

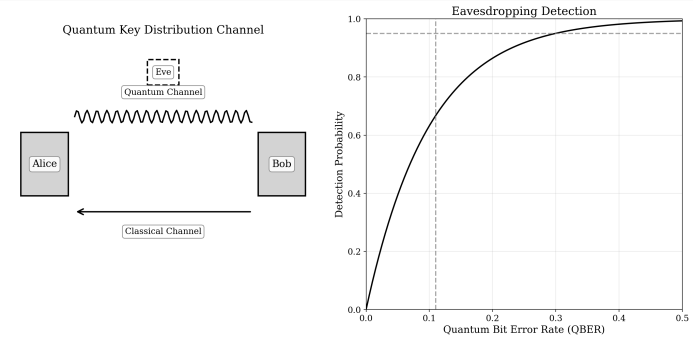

Fig. 3. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: eavesdropping, threshold; The diagram fails because the graph title "Eavesdropping Detection" contains the forbidden word "eavesdropping", which directly reveals a key part of the concept the student is asked to explain.; The diagram fails because the graph title "Eavesdropping Detection" contains the forbidden word "eavesdropping", which directly reveals the concept the student is asked to explain.. A two-part

Question 4

10.0 points

Interpret the no-cloning theorem circuit shown below in the context of quantum key distribution. Explain how this principle prevents cloning of arbitrary quantum states and its implication on secure communication.

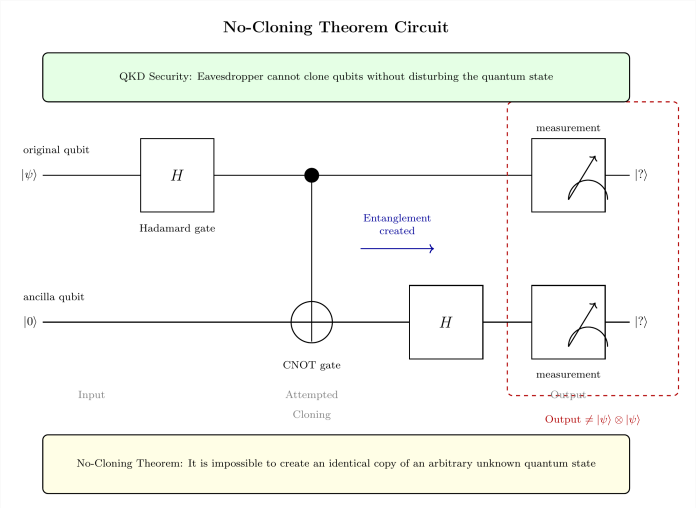

Fig. 4. Quantum circuit diagram illustrating the no-cloning theorem. The circuit includes two qubit lines: one for the original quantum state $|\psi\rangle$ and another for the ancilla qubit initialized to $|0\rangle$. The circuit attempts to clone the state $|\psi\rangle$ using a series of gates, but the no-cloning theorem prevents perfect cloning. Label the qubits as $|\psi\rangle$ and $|0\rangle$, and include a CNOT gate and a Hadamard gate to show the attempted cloning process. Highlight that the output does not result in two identical $|\psi\rangle$ states, illustrating the no-cloning principle.

Question 5 10.0 points

Sketch and analyze the sifted key error rate vs eavesdropping probability curve in the diagram below. Explain the relationship illustrated by the curve and its importance in the BB84 protocol's security strategy.

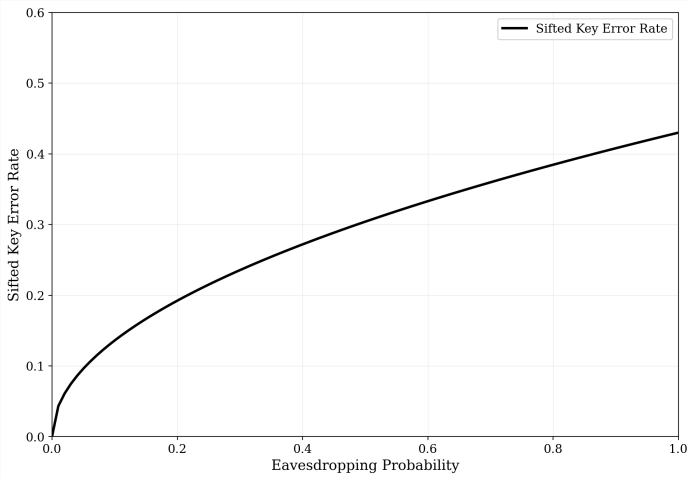

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: eavesdropping, threshold; The diagram includes the forbidden label "Eavesdropping Probability" on the x-axis, which directly reveals a key term the student is expected to use in their analysis.; The diagram fails because the x-axis is labeled "Eavesdropping Probability", which contains the forbidden word "eavesdropping" and directly reveals a key term the student is expected to use in their analysis.. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: eavesdropping, threshold; The diagram includes the forbidden label "Eavesdropping Probability" on the x-axis, which directly reveals a key term the student is expected to use in their analysis.. A plot showing the sifted key error rate vs eavesdropping probability curve for the BB84 protocol. The x-axis represents the eavesdropping probability, while the y-axis represents the sifted key error rate. The curve should illustrate how the error rate increases with higher eavesdropping probability, highlighting the threshold where the error rate becomes significant enough to indicate potential eavesdropping. This visual aids in understanding the trade-off between security and error rate in quantum key distribution. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • eavesdropping • threshold The diagram will FAIL review if any of these appear. IMPORTANT correction needed: The diagram includes the forbidden label "Eavesdropping Probability" on the x-axis, which directly reveals a key term the student is expected to use in their analysis.. Specifically: The x-axis is labeled 'Eavesdropping Probability', which contains the forbidden word 'eavesdropping'.. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT correction needed: The diagram fails because the x-axis is labeled "Eavesdropping Probability", which contains the forbidden word "eavesdropping" and directly reveals a key term the student is expected to use in their analysis.. Specifically: The x-axis label "Eavesdropping Probability. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

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